POLITECNICO MILANO 1863

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IoT Challenge #1

Wokwi and Power Consumption

What to do?

1. Develop a commercial-style IoT motion sensor for a smart home in Wokwi. The sensor must detect the presence of human motion and measure the ambient light level (illuminance), so that a central controller, referred to as the SINK node, can decide whether the room lights should be turned on or off.

2. Implement communication between the motion sensor and a separate central controller. The motion sensor must transmit both the motion status and the ambient light level as a single string, using one of the following formats, where ZZZ is the measured ambient light value:
   - MOTION_DETECTED-LUMINOSITY:ZZZ
   - MOTION_NOT_DETECTED-LUMINOSITY:ZZZ

What to do?

4. After transmitting the message, the motion sensor must enter Deep Sleep for X seconds. When the Deep Sleep interval expires, the node wakes up and resumes its normal operation, which includes sensing and transmission.

5. Estimate the power and energy consumption of one complete operating cycle, and then estimate how long the device can operate using a battery with energy **Y**

6. Discuss the obtained results and propose possible improvements aimed at reducing the overall energy consumption. Support your discussion by implementing the proposed improvements, when possible, and by performing a new energy estimation.

7. Finally, solve the theoretical exercise

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1. IoT Motion Sensor Specification (1.5 pt)

- Design an IoT motion sensor node that communicates to a central ESP32 sink node both the presence of motion and the ambient light level. Based on this information, the sink node can decide whether the lights in the room should be turned on or off.

- In the emulation, use the broadcast address as the ESP32 sink node address: FF:FF:FF:FF:FF:FF

1. IoT Motion Sensor Specification (1.5 pt)

- To reduce energy consumption, the sensor node must operate with a duty cycle. At the end of each round, the node enters Deep Sleep for **X seconds.** After this interval, it wakes up, performs the required sensing operations, and transmits a string in the specified format to the ESP32 sink node. Immediately after the transmission, the node returns to Deep Sleep.
    - MOTION_DETECTED-LUMINOSITY:ZZZ
    - MOTION_NOT_DETECTED-LUMINOSITY:ZZZ
- Communication between the sensor node and the ESP32 sink node must use the ESP-NOW protocol. The sink node is assumed to have no energy constraints and therefore does not need to be implemented.

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1. IoT Motion Sensor Specification (1.5 pt)

- Assume that the ESP32 sink node is always reachable from every sensor node.

The Deep Sleep duration **X**[s], expressed in seconds, must be computed as follows. Take the last two digits of the team leader's person code, compute the value modulo 50, add 5, and then divide the result by 10.

leader_personcode = 106929**AB**

X[s]=(**AB%50+5)/10**

E.g. *leader_personcode* = 10692911

X = (11%50+5)/10 = 1.6 seconds

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2. Energy Consumption Estimation (1 pt)

- Estimate the energy consumption of the sensor node.

  1. Estimate the average **Power** consumption for each state of the node (Deep Sleep state, Idle, Transmission State, Sensor reading, ...)
  2. Estimate the **Energy** consumed during one full transmission cycle.

For these estimations, refer to the power-consumption CSV files provided on Webeep. (The measurements have been described in Lab lecture 2 and for completeness also in the next slides)

2. Energy Consumption Estimation (1 pt)

- To compute the energy consumption of the complete system and to build the power-versus-time graph, you must combine the average power values derived from the CSV files with the timing information obtained from your own implementation in Wokwi.

- The duration of each state can be measured in Wokwi using micros() where possible. Repeated measurements are recommended to reduce variability introduced by the emulator. Make sure the emulator is running (when possible) at around 100%, as indicated by the speed gauge in the top-right corner of the screen.

- Because of Wokwi limitations, the Deep Sleep state cannot be measured reliably during the simulation. For this reason, its duration should be taken from the value set in the code.

2. Energy Consumption Estimation (1 pt)

- Estimate the **operating lifetime** of the sensor node you have implemented in Wokwi, before the battery must be replaced.

Assume that the battery **energy** is **Y** joules, where **Y** is computed from the last four digits of the team leader's person code, according to the following rule:

leader_personcode= 1069**ABCD**

Y=**ABCD%5000+15000 [Joule]**

E.g. code=1069**2911**

Energy = **2911%5000**+15000 = 17911 [J]

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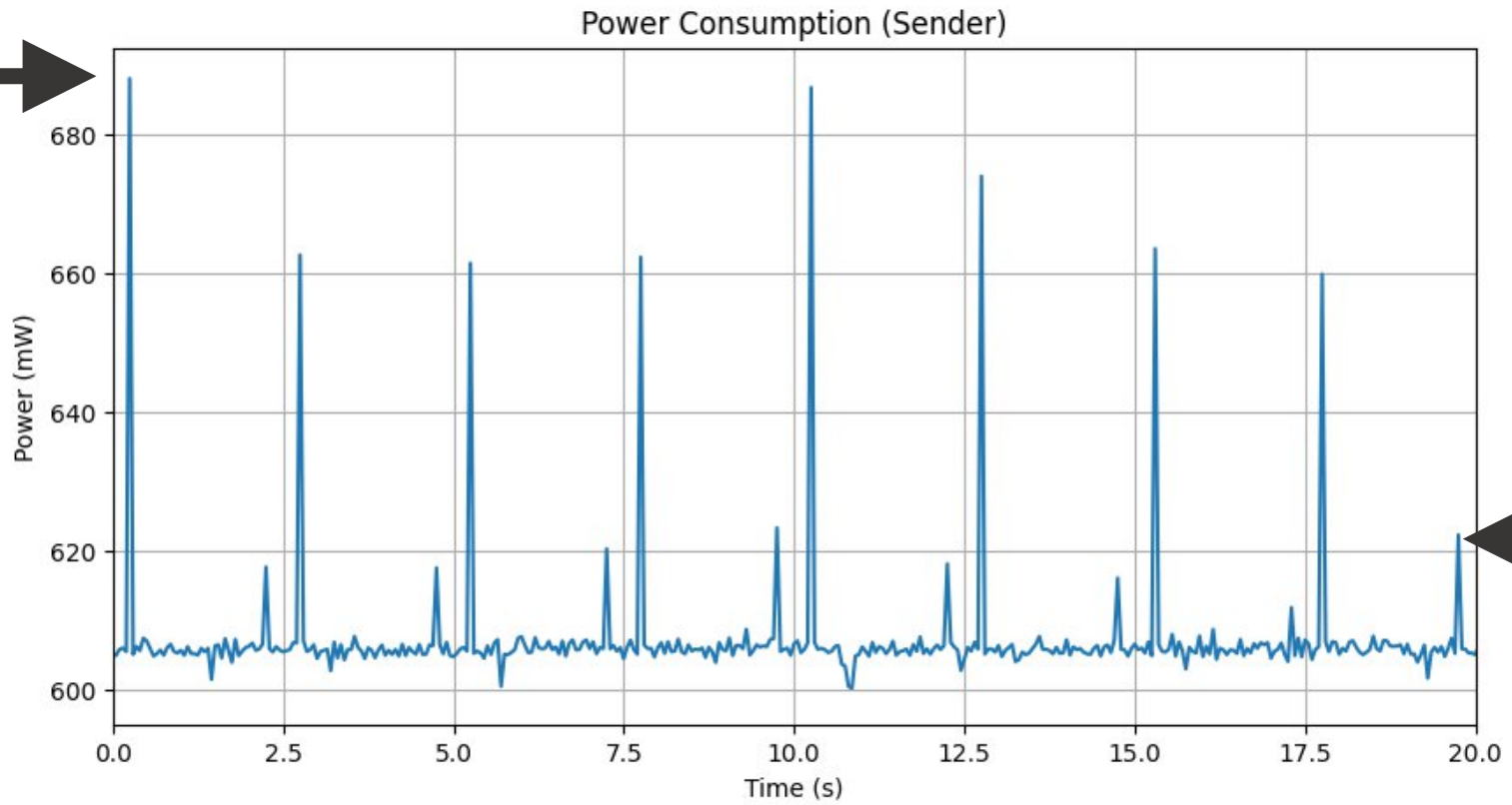
2. CSV File Description

- Each power-consumption CSV file corresponds to a specific execution phase or operating state of the ESP32. Use the provided CSV files to estimate the average power consumed in the different states, including ESP-NOW transmission at different dBm values, a Deep Sleep cycle including idle, Wi-Fi on in RX mode, Wi-Fi off, and Deep Sleep for 10 seconds.

- You are also provided with the power measurements of an ultrasonic distance sensor. For simplicity, use these measurements as a general reference to model the power consumption of a generic sensor-reading phase.

Sender Power measurement

Sending 19.5 dBm
Default value



Sending at 2 dBm

Code Reference

https://wokwi.com/projects/391439340300529665

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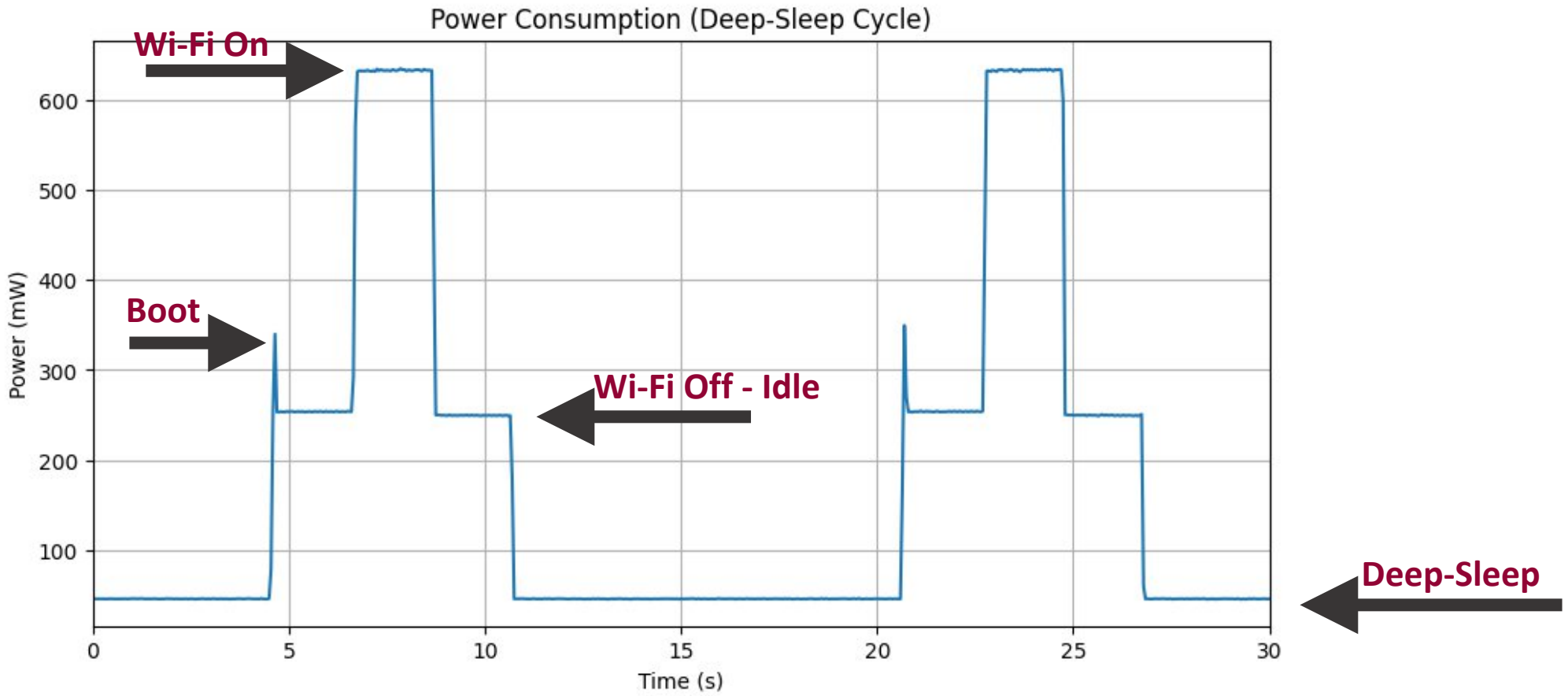
Deep Sleep with Interrupt

Measured Output



Code Reference

https://wokwi.com/projects/389706141289192449

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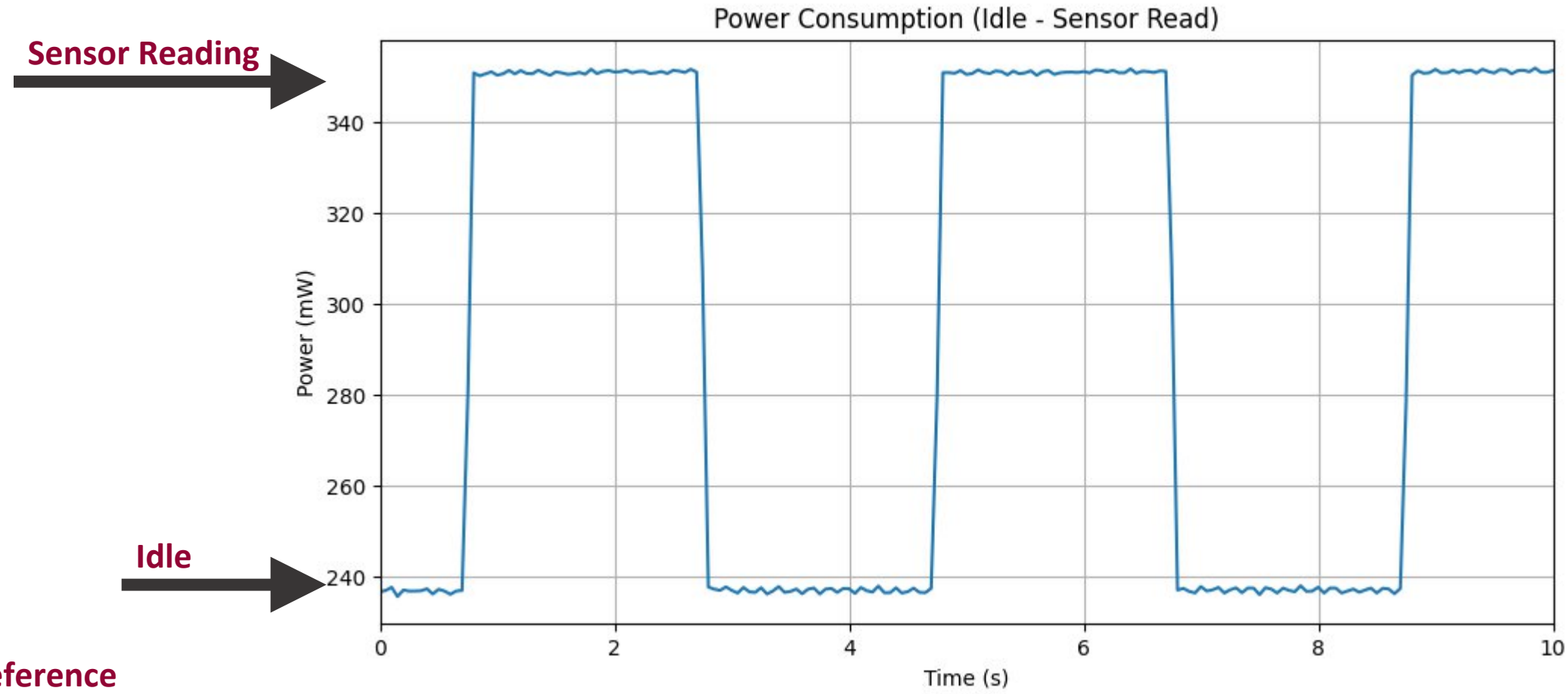
Sensor Reading

Measured Output



Code Reference

https://wokwi.com/projects/458562238151106561

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3. Comment Results and Improvements (2.5 pt)

1. Starting from the system requirements, propose possible improvements aimed at reducing energy consumption without changing the main objective of the sensor node. The goal of the system must remain the same: to notify a sink node about the sensed environmental conditions and support the sink's decision-making process.

2. When possible, reimplement the improved system.

3. Your proposed optimizations must be supported by evidence. Recompute the energy plot after applying the improvements, compare it with the original one, and discuss the differences in terms of energy consumption, design choices, and practical impact.

Evaluation

- The project will be evaluated on the basis of technical correctness, completeness, and the quality of the adopted code logic and design choices.

- The code should be easy to read, properly structured, and must compile and run on Wokwi. Include comments in the code and report it in the final report as well.

- The presentation of the results is also important. Use figures to illustrate power and energy consumption, and discuss the results clearly and thoroughly. Your report should justify the design choices you made, explain the obtained values, and support the discussion in Question 3 with figures and code examples.

Evaluation

- The project will be evaluated on the basis of technical correctness, completeness, and the quality of the adopted code logic and design choices.

For each challenge, a forum page will be available.

Feel free to help one another through the forum. I will also be available there to answer your questions.

- The presentation of the results is also important. Use figures to illustrate power and energy consumption, and discuss the results clearly and thoroughly. Your report should justify the design choices you made, explain the obtained values, and support the discussion in Question 3 with figures and code examples.

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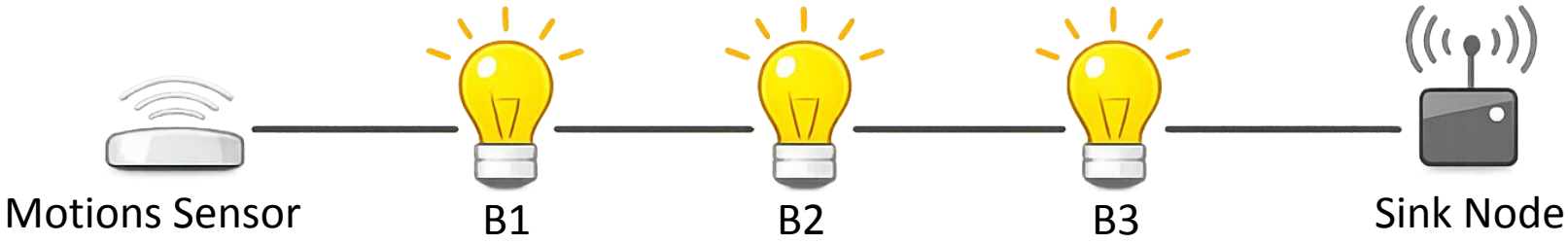
4. Energy Trade-off in a WSN (2pt)

Consider the topology in the Figure:



B1, B2, and B3 are Philips Hue White smart bulbs (see manufacturer specifications)

The presence sensor is an Aqara Motion Sensor P1 (see manufacturer specifications)

The motion sensor has to transmit data to the SINK node. The transmision can occour directly or hop by hop thanks to the bulbs.

4. Energy Trade-off in a WSN (2pt)

The total distance between the presence sensor and the sink is: D = 40 m.

Nodes are equally spaced.

Each transmission of a k = 2000 bit packet follows the energy model:

$E_tx(k,d) = kE_c + k\varepsilon d^2$

$E_rx(k) = kE_c$

where:

$E_c = 300$ nJ/bit

$\varepsilon = 1$ nJ/bit/m^2

Assume ideal communication (no retransmissions, no idle energy).

4. Energy Trade-off in a WSN (2pt)

Q1.

- Compute the **total communication energy consumed by the network** for one message in the case of:
  (a) direct transmission
  (b) multi-hop transmission

Q2.

- Which strategy would you choose for long-term operation?

- Which performance metric are you optimizing when choosing the strategy for long-term operation?

- Provide a brief justification. Your explanation must clearly link the selected strategy to the chosen performance metric and be logically consistent with your numerical results.

Challenge deliverables

What to deliver:

- A **PDF (Named Challenge.pdf)** report containing the explanation of the code logic, the estimation of the power and energy consumption and the reasoning on possible improvements of the sensor node. (Put your NAME and PERSON CODE !!)
  Organize a **clean** report! Very bad reports will be penalized
- **Export of Wokwi project (.ino + .json diagram + libraries used) and/or the shared Public link (when possible) COMMENT AND INDENT YOUR CODE**
- Fill this Form:
  https://forms.office.com/e/h0EUEXEyqh
- Fill this Form to submit the 4th question:
  https://forms.office.com/e/MB65VYWuhb

All the files should be included in a ZIP named as follow:

2-teams: <leader_personcode>_<personcode2>.zip
Single: <leader_personcode>.zip
E.g. 10692911.zip or 10692911_10692912.zip

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For two people teams:

- Choose your team leader and name the file as:
  <leader_personcode>_<other_personcodes>.zip

- **Only the teamleader** should upload the challenge in WeBeep
  Do not upload the same challenge twice

- *Can I take the challenges with the other class students (Prof. Redondi/Cesana)*?
  YES, **but** only the team leader should upload the challenge in WeBeep

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Challenge delivery: HOW?

How to deliver?

- Upload the files in a zip archive as .zip file on the **folder #1** on WeBeep "Assignments" folder

- Fill the two forms:
    - https://forms.office.com/e/h0EUEXEyqh
    - https://forms.office.com/e/MB65VYWuhb

Delivery Deadline

- **STRICT** Deadline:

  March 30, 2026 h 23.59 (FIRM)

- Max **2** people

Good Luck!

WARNING: Code will go thorugh plagarism check, DO NOT COPY !!

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